

AI Automation Basics

Zapier • Make • n8n • Prompt Chaining

No-Code Workflows | Triggers | AI Actions | Engineering Use Cases

What You Will Learn

- What AI automation is and how it differs from simple task automation
- Zapier, Make, and n8n: how they compare and which to start with
- 6 engineering-specific automation workflows you can build today
- Prompt chaining: creating multi-step AI pipelines that pass output to input

Simple Automation Example (Zapier)

Trigger: New email with "invoice" → **Action:** Claude summarises

→ Action: Log amount+vendor to Google Sheet → Action: Telegram alert

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AI-011: AI Automation Basics

You already know how to use AI tools — ChatGPT for calculations, Claude for reports, Perplexity for research. But each of those interactions requires you to open a browser, type a prompt, wait for a response, and then manually do something with the output. That is AI-assisted work. **AI automation** is different: it is AI working continuously in the background, triggered by events, processing information, and producing outputs — without you initiating each step.

The difference between the two is the difference between a calculator (you operate it manually every time) and a spreadsheet with automatic formulas (it updates itself whenever data changes). AI automation makes your AI tools run automatically, triggered by real events in your work, producing real outputs in your existing systems.

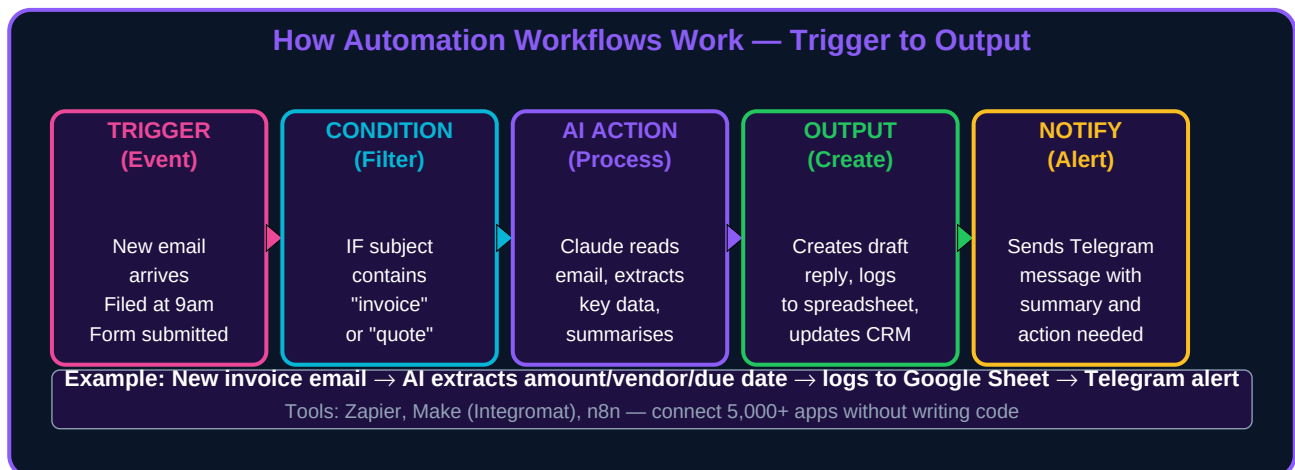
Learning Objectives

- Explain the difference between manual AI use, simple automation, and AI-powered automation
- Compare Zapier, Make, and n8n and select the right tool for your use case
- Build a complete automation workflow using triggers, conditions, AI actions, and outputs
- Design and implement 3 engineering-specific AI automation workflows
- Understand prompt chaining and how to build multi-step AI processing pipelines

1. The Three Levels of Automation

Level	What It Means	Example	Tool
Manual AI	You open a tool, type a prompt, get a response, do something with it manually. Requires your attention each time.	Every morning you open ChatGPT and ask it to summarise your emails	Any AI chatbot
Simple automation (no AI)	Software automatically performs a pre-defined action when a trigger occurs. No intelligence — just rule-based if-then logic.	When a new email arrives, automatically forward it to a folder	Gmail filters, Zapier basic
AI automation	Software monitors for triggers, then invokes an AI model to intelligently process information and take context-aware action — automatically.	When a new email arrives, AI reads it, assesses urgency, drafts a reply, logs key data to a spreadsheet, and sends you a Telegram summary — all while you sleep.	Zapier + Claude API

2. How Automation Workflows Work



Every automation workflow has three core components:

- **Trigger:** The event that starts the workflow — a new email, a form submission, a file appearing in a folder, a scheduled time, a webhook call from another system.
- **Action(s):** What happens in response — send data to an AI model, create a document, update a spreadsheet, send a message, call an API. Most workflows have 3–7 actions in sequence.
- **Output:** The final result — a drafted document, a Telegram notification, an updated database row, a created task in your project management tool.

3. No-Code Automation Tools — Which to Use

No-Code Automation Tools — Zapier vs Make vs n8n

ZAPIER zapier.com	MAKE make.com	n8n n8n.io
6,000+ apps Easiest to use 100 tasks/mo free Best for beginners	1,000+ apps Visual canvas 1,000 ops/mo free Best for complex flows	400+ integrations Open source Free self-hosted Best for developers
~\$19/mo Starter ~\$49/mo Pro Highest cost but easiest	~\$9/mo Core ~\$16/mo Pro Best value mid-tier	Free self-host ~\$24/mo cloud Requires VPS for self-hosting

Feature	Zapier	Make (Integromat)	n8n
Best for	Absolute beginners. Clean, simple interface.	Mid-level users. Complex logic, visual canvas.	Developers or those comfortable with VPS hosting.
Free tier	100 tasks/month, 5 Zaps	1,000 operations/month, 2 active scenarios	Unlimited (self-hosted), limited cloud free
AI integration	Native ChatGPT, Claude, Gemini action blocks	OpenAI, Claude, Gemini modules + HTTP calls	Any AI API via HTTP node. Maximum flexibility.
Visual interface	Simple linear step editor	Full canvas with routers, filters, iterators	Node-based canvas (similar to Make)
Custom code	JavaScript code step available	JavaScript/Python custom functions	Full code nodes — any language
Error handling	Basic retry and error notification	Advanced: error handlers, retry, alternative paths	Advanced: error workflows, full control
Start here if...	You have never built an automation	You have some automation experience	You are comfortable with APIs and servers

Recommended Starting Path

- **Week 1:** Build your first automation in Zapier. Use the free tier. Connect two apps you already use (e.g. Gmail + Google Sheets). This teaches the trigger-action concept without overwhelming you.

- **Week 2–4:** Add an AI action (ChatGPT or Claude step) to your Zapier workflow. Now it is AI automation, not just automation.
- **Month 2:** Move to Make for more complex workflows. Make's visual canvas makes multi-step AI pipelines much easier to design and debug than Zapier's linear format.
- **Month 3+ (optional):** If you want complete control and no per-operation costs, self-host n8n on a \$5/month VPS (DigitalOcean, Hetzner). Unlimited operations for the cost of one server.

4. AI Automation Workflows for Engineers

These six workflows are specifically designed for the tasks engineers do repeatedly. Each can be built in Zapier or Make in under 2 hours with no programming.

6 AI Automation Workflows for Engineers

📧; Email Intelligence

New project email → AI extracts action items → Telegram summary → auto-tags in CRM

📊; Report Generation

Weekly: pull data from Google Sheets → AI writes narrative → email PDF to client

🗣️; Meeting Follow-up

Otter.ai transcript ready → AI extracts decisions/actions → creates tasks in project tool

📄; Document Processing

New invoice uploaded → AI reads PDF → extracts vendor/amount/date → logs to spreadsheet

🚨; Alert Monitoring

Equipment sensor value crosses threshold → AI assesses severity → escalates by priority level

📋; Tender Tracking

New tender published on portal → AI matches to company profile → scores fit → notifies BD team

Workflow Detail — Email Intelligence System

Step	Action	Tool / Module
1. Trigger	New email arrives in Gmail inbox (or Outlook)	Gmail: Watch Emails trigger
2. Filter	Only continue if email is from a client domain OR contains project keywords	Zapier Filter / Make Router
3. AI Extract	Send email body to Claude API: "Extract: (1) sender's request, (2) required action, (3) urgency 1-5, (4) any mentioned deadlines"	OpenAI / Anthropic API action

4. L o g t o s h e e t	Write extracted data to Google Sheets: Date, Sender, Request, Urgency, Deadline	Google Sheets: Add Row
5. D r a f t r e p l y	Send original email + extracted data to Claude: "Draft a professional reply to this request, tone: formal engineering"	OpenAI / Anthropic API action
6. N o t i f y	Send Telegram message with summary + draft reply + link to email	Telegram: Send Message

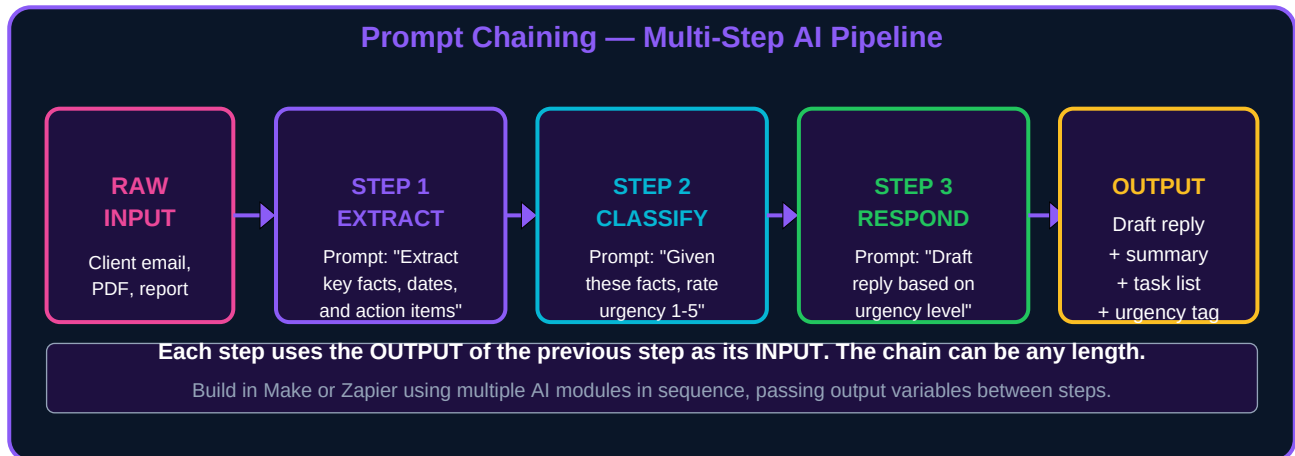
Workflow Detail — Automated Weekly Report

S t e p	Action	Details
1. T r i g g e r	Every Friday at 4:00 PM (scheduled trigger)	Zapier Schedule / Make Scheduler
2. G e t d a t a	Read this week's data from Google Sheets (project hours, tasks completed, issues)	Google Sheets: Get Many Rows

3. A I a n a l y s i s	Send data to Claude: "Write a 200-word weekly project update in professional tone covering progress, risks, and next week's plan"	Anthropic API: Create Message
4. C r e a t e d o c u m e n t	Create a Google Doc with the AI-written report, formatted with your template	Google Docs: Create Document
5. S e n d e m a i l	Email the report link to the project manager and client contact	Gmail: Send Email

5. Prompt Chaining — Multi-Step AI Pipelines

Simple AI automation uses one AI call per workflow. **Prompt chaining** is the technique of connecting multiple AI calls in sequence, where the output of one call becomes the input of the next. This allows you to break complex tasks into focused, reliable steps rather than asking AI to do everything in one prompt (which reduces accuracy and reliability).



Why Chaining Beats Single Prompts for Complex Tasks

Approach	Reliability	Debuggability	Best For
Single long prompt	Lower — AI tries to do too much at once, tends to lose track of earlier instructions	Hard — if output is wrong, unclear which part failed	Simple tasks with one clear output
Prompt chaining (3+ steps)	Higher — each step has a narrow, focused task. AI is better at one thing at a time.	Easy — test each step independently, fix the step that fails without rebuilding the whole chain	Complex multi-output tasks: document processing, report generation, email intelligence

Building Your First Chain in Make

Step	Make Module	Prompt / Action	Output Variable
1	Anthropic: Create Message	"Extract the main request, urgency (1-5), and deadline from this email: {{email_body}}"	{{step1_output}}

2	Anthropic: Create Message	"Given this information: {{step1_output}} — draft a professional response email in under 150 words"	{{step2_output}}
3	Google Sheets: Add Row	Write to sheet: Date Email Subject Urgency Deadline Draft Reply link	N/A
4	Telegram: Send Message	Message: "New client email processed. Urgency: {{urgency}}. Draft: {{step2_output}}"	N/A

6. Getting Started — Your First Automation in 30 Minutes

Minute	Action	What You Are Building
0–5	Create free Zapier account at zapier.com	Your automation account — stores all your workflows (called "Zaps")
5–10	Create new Zap → Choose Trigger: Gmail "New Email" → Connect your Gmail account	The trigger that starts your workflow — any new email in your inbox
10–15	Add Action: "ChatGPT" → Connect OpenAI account (or use Zapier's built-in) → Choose "Conversation"	The AI step that processes the email content
15–20	In the ChatGPT prompt field, type: "Summarise this email in 2 sentences and state the required action: {{Body Plain}}"	Your first prompt — uses the email body as dynamic input
20–25	Add second Action: Gmail "Send Email" → To: yourself → Subject: "AI Summary" → Body: {{ChatGPT Response}}	The output step — sends you the AI summary by email

25–30	Test the Zap → Turn it ON → Send yourself a test email → Check for the AI summary reply	Your first working AI automation — runs automatically on every new email from now on
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7. Practice Exercises

#	Exercise	Instructions
1	Build your first Zap	Create a Zapier account (free). Build a Zap that: (a) triggers on a new Gmail email, (b) sends the email body to ChatGPT with the prompt "Summarise this email in 3 bullet points", (c) sends you the AI summary via Gmail to your own address. Turn it on and test with a real email.
2	Add a filter	Edit your Zap from Exercise 1. Add a Filter step between Gmail and ChatGPT that only continues if the email subject contains "project" or "urgent". Test with two emails — one that should pass the filter and one that should not. Verify only the correct email triggers the AI step.
3	Build in Make	Create a free Make account (make.com). Recreate your Gmail → AI summary workflow in Make. Notice how Make's visual canvas shows the data flowing between modules. Add a Google Sheets step that logs: date, sender, AI summary to a spreadsheet.
4	Engineer your own workflow	Identify ONE repetitive task in your daily engineering work. Map it as: Trigger → Filter → AI Action → Output. Build it in Zapier or Make. Document the exact prompt you use and the result quality. Refine the prompt until the output is professional-quality.
5	Build a prompt chain	In Make: create a 3-step prompt chain for processing a project brief. Step 1: "Extract: project scope, deliverables, timeline, and constraints from this text". Step 2: "Based on these project details, identify the top 5 risks". Step 3: "Write an executive summary paragraph covering scope, timeline, and key risks". Test with a real or sample project brief.

Lesson Summary

Concept	Key Rule
AI automation vs manual AI	Manual: you operate AI each time. Automation: AI runs triggered by events, without your intervention.
Workflow components	Every workflow has: Trigger (what starts it) + Action(s) (what happens) + Output (the result).
Tool selection	Beginners: Zapier. Intermediate: Make. Developers: n8n (self-hosted). All can call AI APIs.
Prompt chaining	Multiple AI calls in sequence, passing output from one to input of next. More reliable than single long prompts.
Best first automation	Email intelligence: new email → AI extracts key info → logs to sheet → Telegram alert. Saves 30+ min/day.
Building correctly	Start with one simple Zap. Add AI after it works without AI. Then add complexity. Never build everything at once.

What Is Next — AI-012

AI-012: AI Agents Introduction — What AI agents are and how they differ from simple chatbots. Agentic AI systems that plan, use tools, and complete multi-step tasks autonomously. Claude Computer Use, OpenAI Operator, and Auto-GPT. Building your first agent workflow with real-world tools. Safety considerations for autonomous AI systems.

Resource	Details
AI PDF Series	AI-001 through AI-030 — ayetechub.com/pdfs.html
AI Tools Directory	Zapier, Make, n8n reviewed — ayetechub.com/ai-tools.html
Community	Telegram: t.me/ayetechub